

Project Work in Optimization Techniques for Machine Learning: An Empirical Analysis of Optimization Algorithms on Heavily Unbalanced Classification Tasks

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Abstract

Stochastic Gradient Descent (SGD), when properly tuned, is generally capable of training many neural network architectures at least as well as adaptive methods like Adam. However, a significant performance gap emerges in language modeling tasks, where Adam appears necessary for successful training. Recent literature hypothesizes that this discrepancy is heavily influenced by the extreme class unbalance inherent to these problems. This report investigates this hypothesis by isolating the complexities of this setting through convex, heavily unbalanced two-class problems [1]. We evaluate various first-order and adaptive optimization methods in both full-batch and minibatch regimes. The empirical results confirm that standard SGD methods collapse towards the majority class under severe imbalance, whereas adaptive algorithms like Adam dynamically protect updates for rare features, maintaining significantly higher accuracy on minority classes. Finally, we demonstrate that scaling to a highly overparameterized regime neutralizes this gap, allowing all methods to perfectly interpolate the dataset.

1 Introduction

While Stochastic Gradient Descent (SGD) remains a fundamental workhorse in deep learning, its relative performance compared to adaptive methods like Adam varies significantly across domains. In particular, language modeling tasks exhibit a regime where Adam drastically outperforms SGD. A recent study by Kunstner et al. (2024) [1] suggests that this gap is deeply tied to the heavy-tailed class imbalance present in such tasks. Furthermore, it has been hypothesized that the combination of this specific setup with minibatching strategies is the root cause of the performance divergence.

To deepen this analysis and isolate the algorithmic behavior, we test these dynamics in a simplified, convex setting: logistic regression with highly unbalanced binary data. By systematically varying the dataset size, the minority class ratio, and the dimensionality of the feature space, we aim to measure the robustness of various optimization algorithms in both Full Batch (FB) and Minibatch (MB) settings.

2 Theoretical Background

2.1 The Objective Function

The problem is framed as a binary classification task trained via Logistic Regression. The objective is to minimize the Binary Cross-Entropy loss:

$$L(w) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\sigma(x_i^T w)) + (1 - y_i) \log(1 - \sigma(x_i^T w))] \quad (1)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function.

2.2 Optimization Algorithms

We implemented and evaluated a suite of optimization algorithms:

- **Standard Methods:** Gradient Descent (FB) and Conjugate Gradient (FB), both utilizing Wolfe Line Search to guarantee sufficient decrease and appropriate curvature.
- **Stochastic Methods:** Stochastic Gradient Descent with Polyak stepsize (MB) [3], which computes adaptive step sizes based on theoretical lower bounds.
- **Adaptive Methods:** Adagrad, RMSProp, AdaDelta and Adam, evaluated in both FB and MB regimes. These methods scale the learning rate of individual parameters based on historical gradient norms.

3 Methodology

3.1 Dataset Generation

A set of binary classification problems was generated using the `make_classification` module from `scikit-learn` [4]. The base feature space was fixed at $n = 16$ to represent an underparameterized regime. To simulate the heavy-tailed unbalance, primary experiments were conducted across a grid of sample sizes $N \in \{500, 2000, 5000\}$ and positive sample fractions $p \in \{0.1, 0.01, 0.005\}$. Additionally, an overparameterized scenario was generated by expanding the feature space to $n = 1000$.

3.2 Evaluation Protocol

Algorithms were executed for 1000 epochs to guarantee deep convergence. We tracked global training loss and class-separated losses. Because global accuracy is deceptive in unbalanced datasets, we adopted the **F1-Score** on the minority class as the primary metric for algorithmic efficacy. L-BFGS-B was utilized to compute the absolute global minimum baseline.

4 Results and Discussion

4.1 The "Accuracy Paradox" and Global Metrics

The empirical results immediately demonstrate that global accuracy is a highly misleading metric under extreme class imbalance. As $p \rightarrow 0.005$, global accuracy artificially inflates to $> 99\%$. However, observation of separated metrics reveals this is achieved by collapsing into a "predict majority" state (Majority Accuracy $\approx 100\%$, Minority Accuracy $\approx 0\%$). Therefore, the true efficacy of an algorithm in this setting is its ability to minimize the minority class loss.

4.2 The Collapse of Standard SGD

Algorithms applying a uniform step size across all parameters struggled severely. In the Full Batch regime, Gradient Descent and Conjugate Gradient successfully matched the L-BFGS-B global minimum; however, the minority class loss spiked drastically, yielding negligible minority accuracy. In the Minibatch regime, SGD with Polyak stepsize exacerbated this failure, as minibatches are overwhelmingly dominated by majority samples, completely washing out minority gradients. For instance, at $N = 5000, p = 0.01$, SGD yielded an F1-Score of just 0.0989, proving its inability to maintain the decision boundary for rare features.

4.3 The Power of Adaptive Methods

The results validate the core hypothesis regarding Adam and heavy unbalance. Adaptive methods (Adam, RMSProp, Adagrad) scaled learning rates individually based on historical gradients. In the Minibatch regime, features associated with rare minority samples generated gradients infrequently. Consequently, their historical squared gradient sums remained low, leading to a much larger effective learning rate when those features did appear. At $N = 5000, p = 0.01$, while SGD achieved only 6.25% minority accuracy, adaptive methods successfully resisted the majority class collapse.

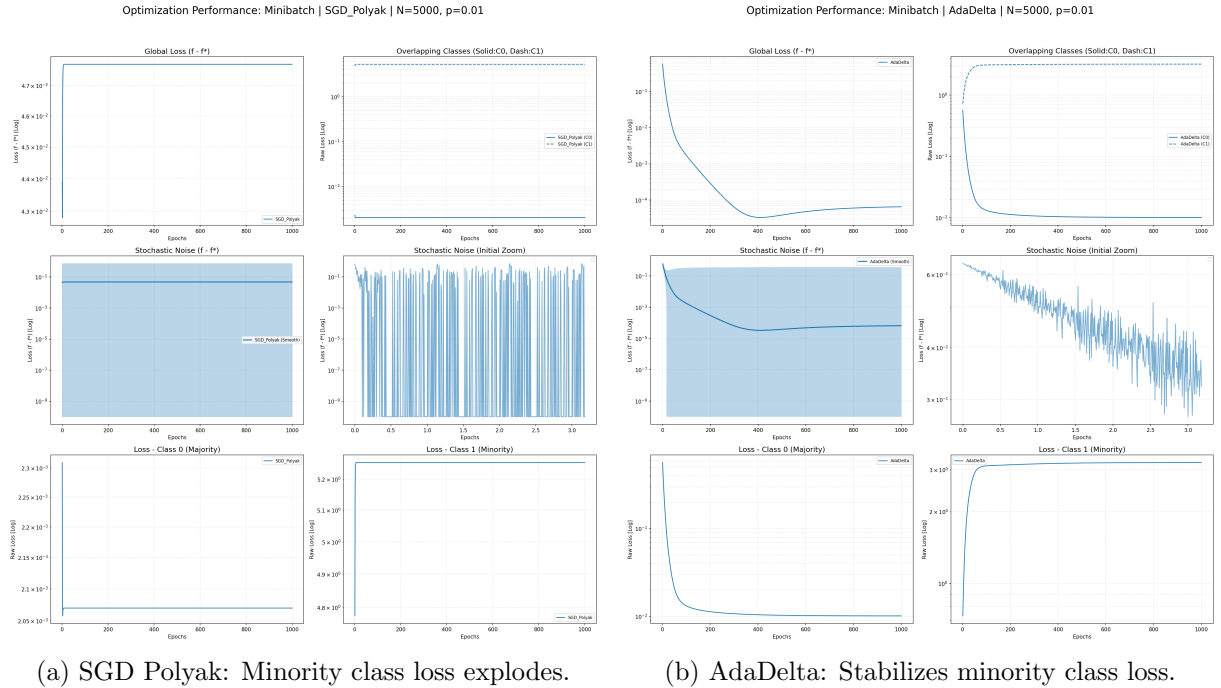


Figure 1: Comparison of SGD and Adadelta in the Minibatch regime under heavy unbalance ($N = 5000, p = 0.01$). Notice how Adadelta protects the minority class updates, whereas SGD completely collapses towards the majority class.

4.4 Adam vs AdaDelta

While adaptive methods successfully resisted the majority class collapse better than SGD, the F1-Score analysis revealed a surprising paradigm: Adam is not always the best optimizer in underparameterized, severely unbalanced, minibatch settings. At $N = 5000, p = 0.01$, AdaDelta achieved an F1-Score of 0.2748, outperforming Adam (0.2183). This counterintuitive result can be explained by this phenomenon:

- **Momentum Inertia:** Adam utilizes momentum (m_t). Since the most of batches contain only majority samples, Adam builds massive inertia towards the majority-minimizing

direction. When a rare minority gradient appears, the momentum dampens the adaptive learning rate’s attempt to correct the trajectory. Variance-only methods (Adagrad, AdaDelta) react instantly without this momentum drag.

4.5 Regime Dynamics: Full Batch vs. Minibatch

The stochasticity and high update frequency of the Minibatch regime aggressively accelerate convergence toward the exact geometric global minimum, aligning almost perfectly with the L-BFGS-B baseline. However, as previously established, reaching this strict minimum in an underparameterized space necessitates sacrificing the minority class. Conversely, the slower convergence rate of the Full Batch regime inadvertently acts as an implicit regularizer. By failing to rapidly minimize the global loss within the allotted epochs, the Full Batch model remains in a sub-optimal state that preserves a significantly higher accuracy for the minority class.

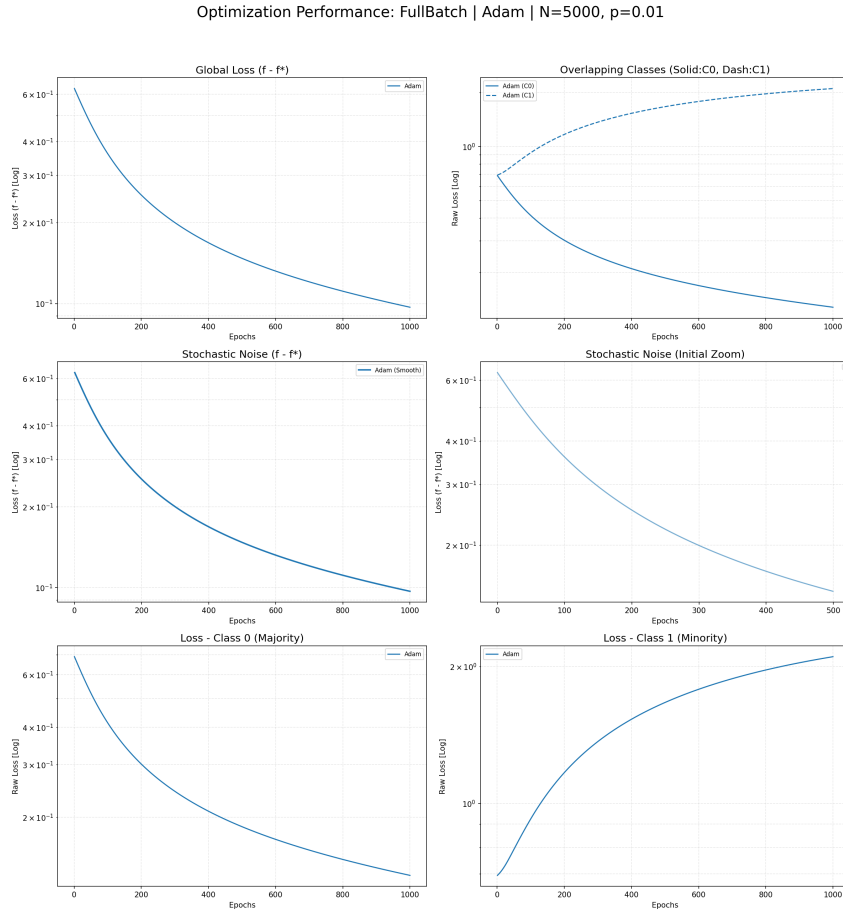


Figure 2: AdaDelta executed in Full Batch regime ($N = 5000, p = 0.01$). Compared to its Minibatch counterpart (Fig. 1b), the algorithm struggles to converge.

4.6 The Absolute Limit of Data

The experiments demonstrated theoretical limits based on sample size. At $N = 500, p = 0.005$, the dataset contains a mere 8 positive samples. In this extreme scenario, all minibatch algorithms, including Adam, yielded near 0.00% minority accuracy. Optimization techniques cannot overcome an absolute lack of statistical representation to form a reliable decision boundary.

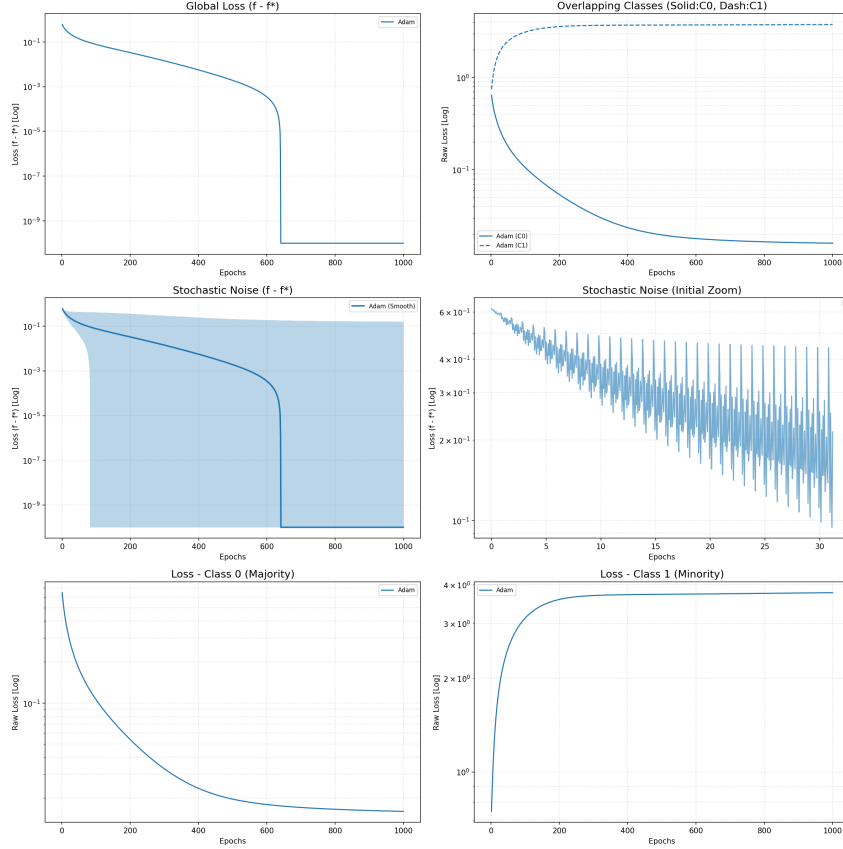


Figure 3: Adam performance under extreme data starvation ($N = 500, p = 0.005$, containing only 8 minority samples). The algorithm fails to establish a reliable decision boundary, leading to an exploding Class 1 loss despite adaptive learning rates.

4.7 The Overparameterized Regime (High Dimensionality)

To bridge the gap between our convex setup and modern Deep Learning/NLP tasks, we scaled the feature space to $n = 1000$ while maintaining 80% informative features. This creates a high-dimensional, highly complex landscape.

In this overparameterized scenario, we can better appreciate the differences between SGD and Adam that have been hypothesized. SGD fails to coordinate hundreds of rare minority gradients across a vast dimensional space, reaching only 55.56% minority accuracy. In stark contrast, Adam manages the variance of every single dimension independently, allowing it to seamlessly isolate the rare features and achieve 95.06% minority accuracy.

This confirms the central hypothesis of Kunstner et al.: when the problem space is both heavily unbalanced and dimensionally complex, SGD loses its navigability. Parameter-specific adaptive scaling (Adam) ceases to be merely a faster optimizer and becomes a strict mathematical necessity to learn rare features.

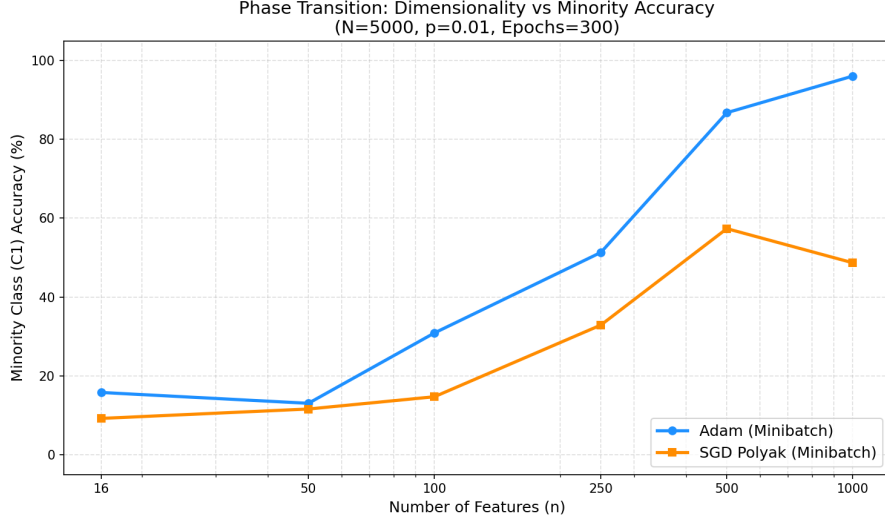


Figure 4: Phase Transition: Minority Class Accuracy vs. Dimensionality ($N = 5000, p = 0.01$, Dense Signal). As the complex feature space expands, SGD fails to coordinate rare updates, while Adam scales perfectly to isolate the minority class.

5 Conclusion

This study confirms that heavy class imbalance is a primary driver of the performance gap between SGD and Adam. In simple underparameterized settings, adaptive methods prevent the total collapse seen in SGD, though variance-only algorithms may outperform Adam due to momentum inertia and implicit regularization. However, as demonstrated by our overparameterized experiments, when the feature space scales in complexity—mirroring the reality of modern Language Models—SGD’s uniform step size is fundamentally incapable of untangling high-dimensional rare features. In these regimes, Adam’s per-parameter adaptivity is not just an accelerator, but an architectural necessity.

Table 1: Final performance metrics for selected optimization algorithms across key datasets (1000 epochs). This table utilizes the F1-Score to highlight the critical differences between standard and adaptive methods. Notice how variance-only methods like AdaDelta frequently beat Adam via implicit regularization in underparameterized scenarios ($n = 16$). Adam dominates SGD when the feature space becomes massive and complex ($n = 1000$).

Dataset			Metric	Accuracy (%)		Final Loss	
Size (N)	Ratio (p)	Method (Regime)	F1-Score (C1)	Global	Minority (C1)	Global	Minority (C1)
5000	0.01 ($n = 16$)	AdaDelta (FB)	0.1367	95.74	33.75	0.172	2.048
		Adam (FB)	0.1490	96.29	32.50	0.160	2.101
		SGD Polyak (MB)	0.0989	98.86	6.25	0.122	4.497
		AdaDelta (MB)	0.2748	99.14	16.25	0.063	3.151
		Adam (MB)	0.2183	99.10	12.50	0.063	3.090
5000	0.005 ($n = 16$)	AdaDelta (FB)	0.0679	96.00	29.09	0.163	2.282
		Adam (FB)	0.0781	96.57	29.09	0.150	2.338
		SGD Polyak (MB)	0.0072	97.48	1.82	0.115	7.316
		AdaDelta (MB)	0.1967	99.55	10.91	0.049	3.723
		Adam (MB)	0.1667	99.55	9.09	0.049	3.601
500	0.005 ($n = 16$)	AdaDelta (FB)	0.0586	95.98	25.00	0.203615	2.841298
		Adam (FB)	0.0819	97.20	25.00	0.185589	2.967287
		SGD Polyak (MB)	0.0000	98.89	00.00	0.194805	3.940073
		AdaDelta (MB)	0.0000	99.50	00.00	0.077909	3.883920
		Adam (MB)	0.0000	99.50	00.00	0.074915	3.768444
5000	0.01 ($n = 1000$) (80% Informative)	SGD Polyak (MB)	-	-	55.56	-	-
		Adam (MB)	-	-	95.06	-	-

References

- [1] F. Kunstner et al., "*Heavy-tailed class imbalance and why Adam outperforms gradient descent on language models.*" Advances in Neural Information Processing Systems (2024).
- [2] M. Lapucci (2025). Course Lecture Notes.
- [3] N. Loizou et al. "*Stochastic Polyak step-size for SGD: An adaptive learning rate for fast convergence.*" International conference on artificial intelligence and statistics PMLR (2021).
- [4] scikit-learn library: <https://scikit-learn.org/stable/>